Clustering

Clustering

- an answer to unavailability of labeled data – no characteristics of the groups (classes) is known
- goal = grouping data into subsets called clusters using information contained *in the data*
- the clusters are *coherent* (homogeneous) internally and must be clearly *different* from each other
- no labels = no one correct solution

Clustering applications

- grouping/labeling large amounts of unlabaled data
- navigating the results retrieved by a search engine, cluster pruning
- text summarization
- topic detection

What to cluster

- document clustering is the most common, e.g., to discover topics or organize a collection of documents
- shorter pieces (e.g., sentences) can be clustered too (e.g., in extractive summarization, one sentence from a cluster can be selected)
- word clustering can help in reducing dimensionality (words in the same cluster are replaced by one variant)

Clustering algorithms

- *hard* clustering – each object belongs exactly to one cluster
- *soft* or *fuzzy* clustering – each object is associated with each cluster with a degree of membership in the interval [0, 1]
- different clustering algorithms usually lead to different clustering solutions
- clustering optimization is a very hard task (e.g., for a hard, flat clustering of n elements into k clusters there exist k^n possible solutions)

Clustering algorithms

- *partitional (flat) clustering* seeks a k-partition

  $C = \{C_1, C_2, \ldots, C_k\}, k \leq n$ of X

  such that

  $C_i \neq \emptyset, i = 1, 2, \ldots, k;$

  $\bigcup_{i=1}^{k} C_i = X;\ C_i \cap C_j = \emptyset, i, j = 1, 2, \ldots, k \text{ and } i \neq j$

Clustering algorithms

- *hierarchical clustering* constructs a tree like, nested structure partition of X,

  $$H = \{H_1, H_2, \ldots, H_k\},\ k < n,$$

  such that $C_i \in H_m$, $C_j \in H_l$, and $m > l$ imply $C_i \subset C_j$ or $C_i \cap C_j = \varnothing$ for all $i, j \neq i, m, l = 1, 2, \ldots, k$

Clustering algorithms

- *graph clustering* – partitioning a *similarity graph* (vertices = documents, edges = quantified according to the similarity between the documents), i.e., minimization of the edge-cut; an alternate model considers the documents and the terms contained in them to be the vertices and weights of the edges are set using *tf-idf* measure

Partitioning clustering

- no explicit structure between the clusters is considered
- their result is a set of *k* clusters, where *k* is given or automatically determined
- well suited for clustering large document data sets due to relatively low computational requirements

K-means

- the most widely used flat clustering algorithm
- in the first step, *k* cluster centers are randomly selected
- then, all objects are assigned to a cluster with the closest center
- in the following step, the cluster centroids are re-computed according to the positions of the objects in the clusters
- the steps of assignment of the objects to the clusters and re-computation of cluster centroids are repeated until a stopping criterion has been met
- the algorithm might arrive to a local optimum – the entire process might be repeated several times with a different seed, outliers might be excluded from the seed set, or the seed might be obtained from another method

K-means modifications

- K-medoids
  - an object that is closest to the centroid is used as a representative
  - suitable when many vector values are zeros (e.g., clustering documents using the tf-idf representation)
  - only nonzero features participate in distance calculations (decreasing computational complexity)

Optimizing criterion function

- proposed by Zhao and Karypis (2005), recommended for document clustering
- an initial clustering solution is refined in iterations in order to improve the value of a given criterion function

Optimizing criterion function

1. k documents are selected as cluster centers
2. other documents are assigned to clusters
3. for each document x (visited in a random order) :
    a) the value of the criterion function c is calculated
    b) x is assigned to all other clusters and the value of *c* is calculated for each such assignment
    c) x is moved to the cluster that improves the value of c the most
4. if some documents have been moved, go to step 3

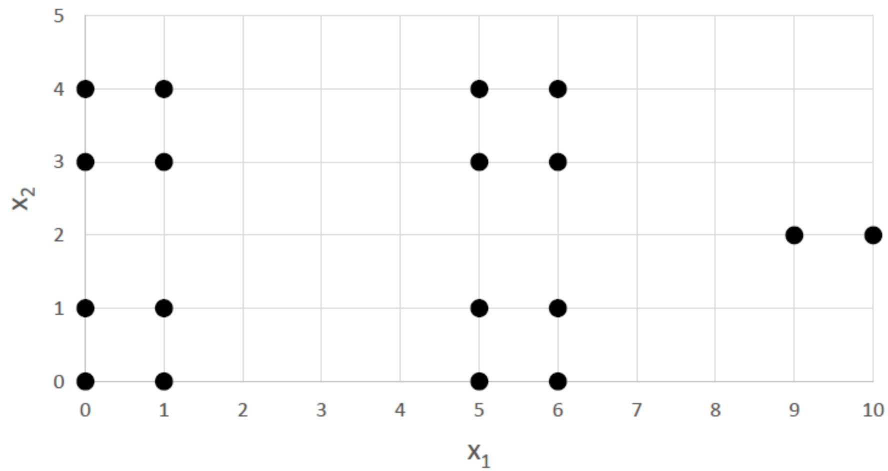
Optimizing criterion function

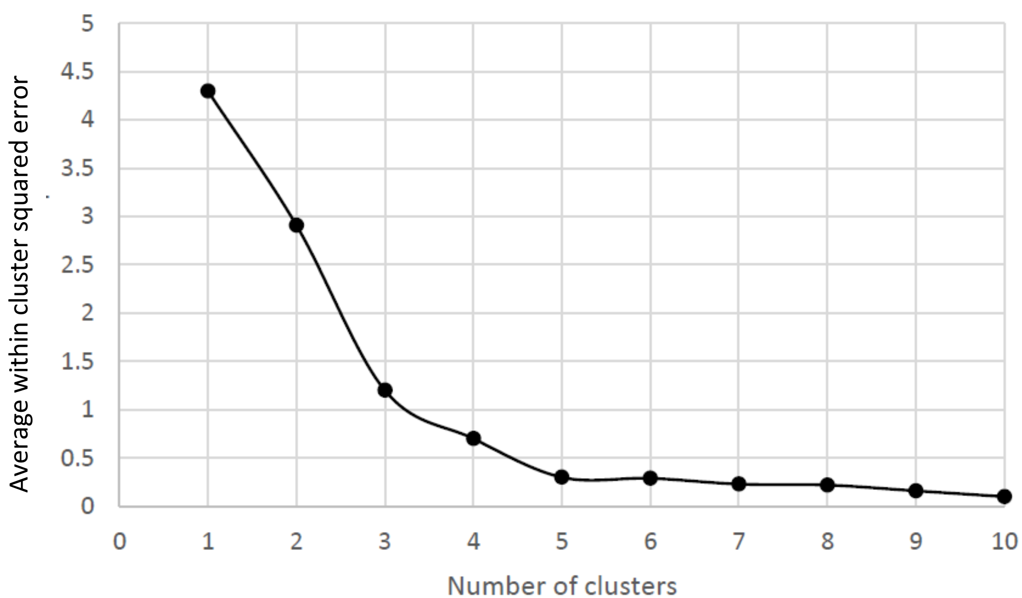
- internal criterion functions
  - I1: maximizes the sum of average pairwise similarities between the document assigned to each cluster, weighted according to the size of each cluster
  - I2: maximizes the similarity between documents and the centroids of the clusters
- external criterion function
  - E1: maximizes the distance between cluster centers and the centroid of the entire dataset
- hybrid criterion functions
  - combine I1, I2, and E1 (H1=I1/E1, H2=I2/E1)

The number of clusters

- important parameter mainly for flat methods
- usually not known
- usually only a human expert can decide on the usefulness of clustering

The elbow method – example





The elbow method – example

- we have five clearly visible clusters
- the objects are clustered to 1 to 10 clusters using k-means with the Euclidean distance
- the average within cluster squared error was calculated for each solution (any other function measuring the quality clustering can be used)
- when each object is in a separate cluster, the average error is zero (i.e., the global minimum); the clustering is, however, not useful

The elbow method

- at the beginning, the error function value decreases quite significantly; the sharp decrease stops when the number of clusters is five (we can see an *elbow* or a *knee*)
- five clusters is a good compromise between the quality of clustering and the generality
  – more clusters will not improve the solution (measured with the error) considerably
  – a smaller number of clusters will increase the error significantly

The number of clusters

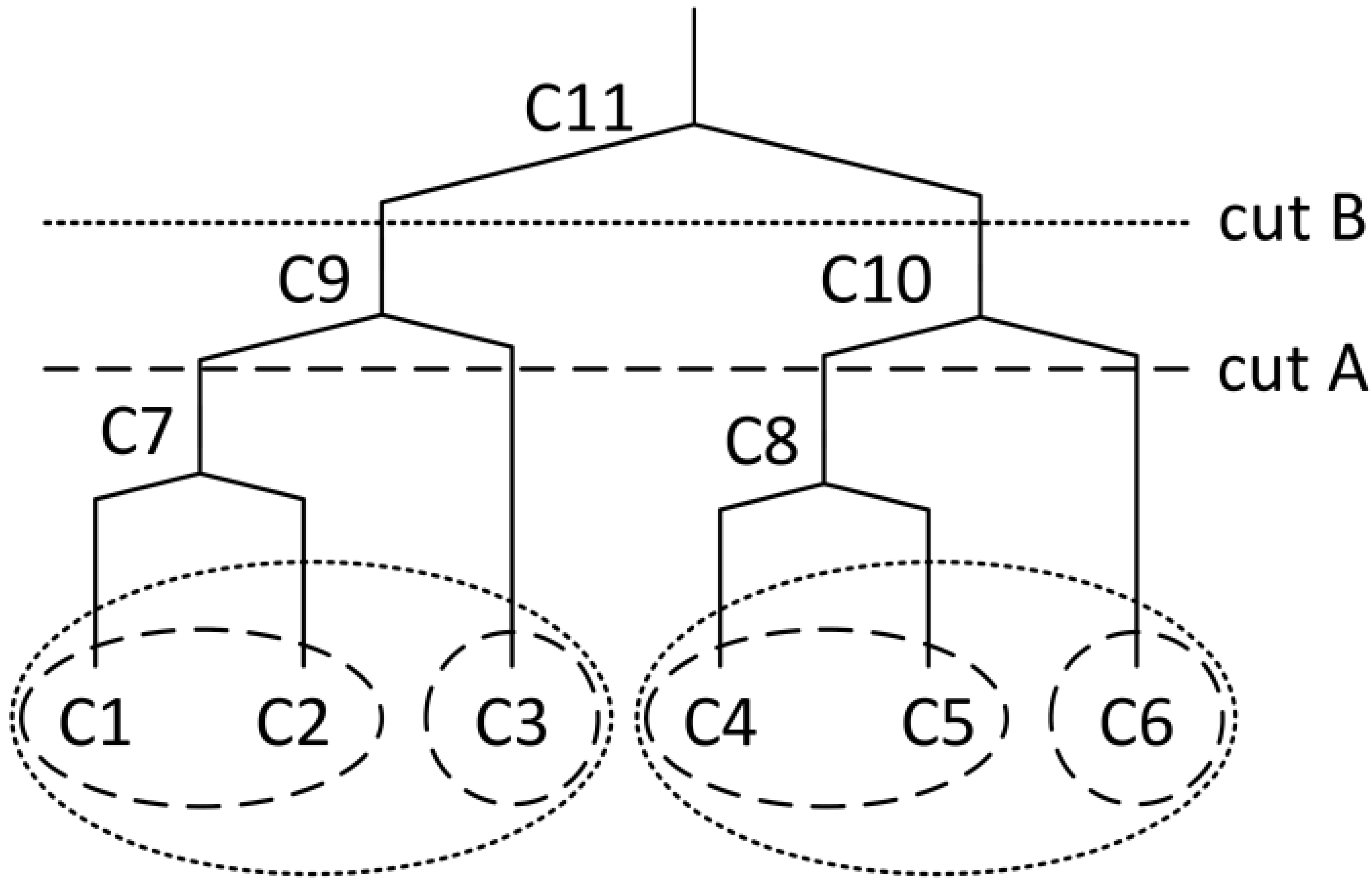
- another metric can be optimized, e.g., the silhouette is maximized
- a recommendation related to a specific task can be used; for example, in information retrieval, the number of clusters can be m*n/t, where n is the number of terms, m the number of documents, and t is the total number of term assignments

Hierarchical clustering

- top-down (divisive) – at the beginning is all data in one cluster; in the following steps, the data is split into several clusters (typically using a flat method); the same step is then recursively repeated for the new clusters according to some criteria
- bottom-up (agglomerative) methods – each item to be clustered is in a single cluster at the beginning; in each of the following steps, two most similar clusters are joined together until all objects are in one cluster

Merging clusters in agglomerative methods

- *single linkage clustering method* (SLINK), the clusters are merged according to the similarity of the most similar objects from the clusters
- *complete linkage clustering method* (CLINK), determines the similarity according to the similarity of two most dissimilar objects
- *Unweighted Pair Group Method with Arithmetic Mean* (UPGMA), the distance between two clusters is calculated as the average of all distances between pairs of objects
- *Unweighted Pair Group Method with Centroid (UPGMC)* – instead working with all objects, the means are used in calculations



A dendrogram as a result of agglomerative hierarchical clustering. By performing different cuts different clusters are obtained.

Divisive hierarchical clustering

- construct the hierarchy in the top-down manner
- all objects are initially in one cluster which is recursively divided until we have only singleton clusters
- to obtain the same type of hierarchy as in case of agglomerative clustering, the cluster is usually divided into two sets

Divisive hierarchical clustering

- it is often not necessary to build a complete dendrogram – divisive clustering is more efficient than agglomerative
- the stopping criteria may include
  - k clusters are found
  - maximal hierarchy depth
  - minimum number of objects in a cluster
  - minimal homogeneity (dissimilarity) in a cluster

Divisive hierarchical clustering

- choosing a cluster to split
  - the larger cluster is divided
    - usually leads to reasonable good and balanced clustering solutions
    - not suitable when there naturally exist large clusters
  - the cluster is split so the value of a selected criterion function improves the most

Constrained clustering

- might use constraints imposed by a flat clustering that initially partitions the data
  - it is not possible to merge objects or clusters from different constraint clusters
  - combines the global view of flat methods and local view of agglomerative methods
  - eliminates the problems of agglomerative clustering (local perspective), the time complexity decreases

Constrained clustering

- another possibility to change the unsupervised process of clustering is to bring some external knowledge in the form of
  - must-links (information about objects that must be in the same cluster)
  - cannot-links (information about objects that cannot be in the same cluster)
  - COP-K-means
    - modifies the k-means algorithm
    - an object is assigned to the corresponding cluster only if a cluster satisfying the constrains is found

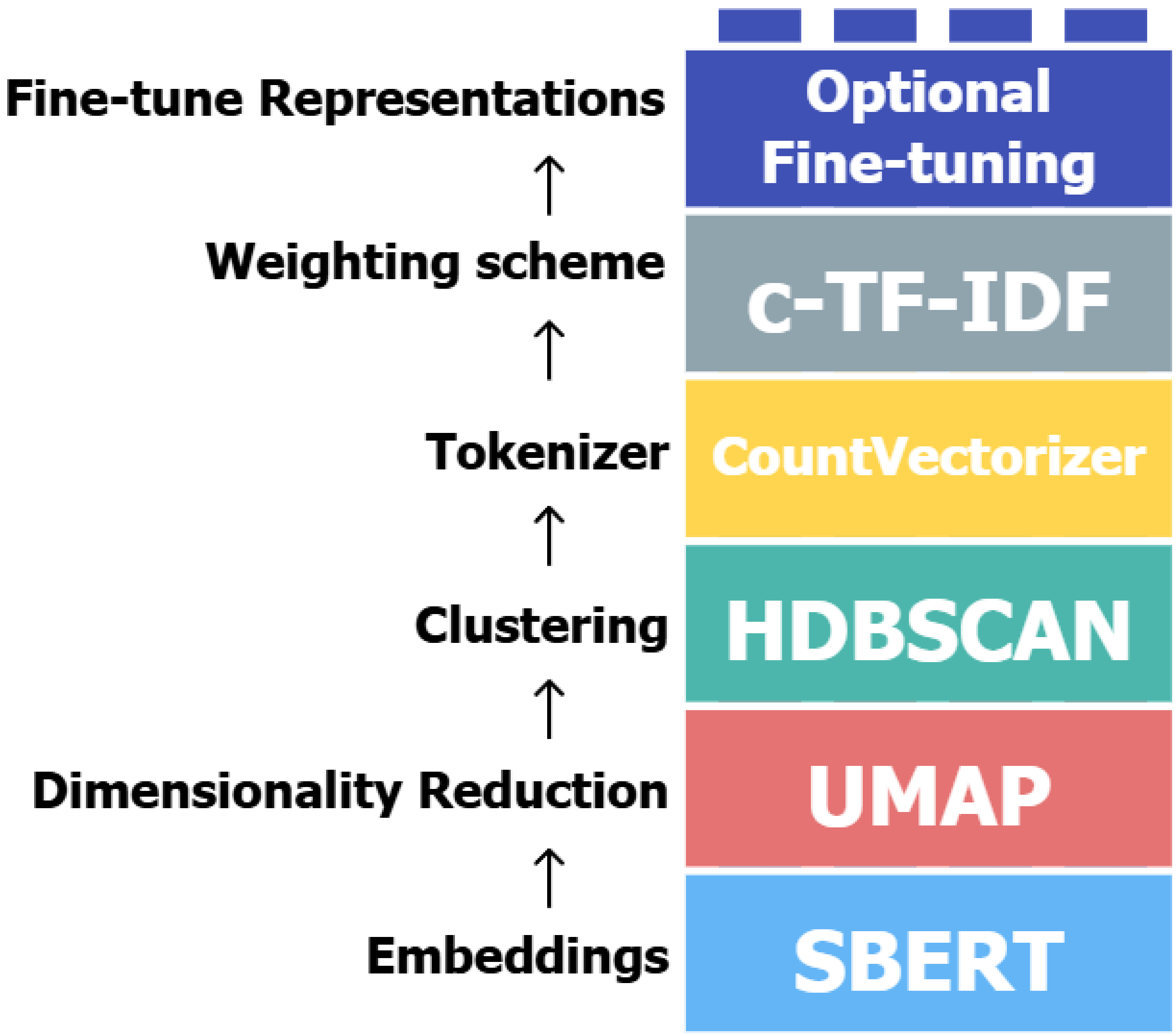
Modern approaches

- LLMs can provide better representations based on embeddings
  - smaller, dense representations
  - better semantics handling
- LLMs can interpret clusters, label them, summarize them, …

BERTtopic

- converts documents to embeddings using an LLM (trained for semantic similarity might be a good idea)
- projects the embeddings to fewer dimensions (typically with UMAP)
- the documents are clustered (e.g., HDBSCAN)
- documents from each cluster are merged into one -> bag-of-words representation on the cluster level (TF*IDF/sum(TF) – class-based TF-IDF)
- words with the highest c-tf-idf scores describe the topic
- modular approach, each level can be done alternatively

https://arxiv.org/abs/2203.05794



https://maartengr.github.io/BERTopic/algorithm/algorithm.html

Similarity

- the only endogenous information that is available in the clustering process, used also by instance based classifiers (k-NN)
- can be measured between individual documents, between groups of documents, or between a document and a group of documents
- a vector representing a group of documents

$$C_C = \frac{\sum_{x \in C} x}{|C|} = (C_{C1}, C_{C2}, \ldots, C_{Cm})$$

Similarity measures

- cosine similarity
  - based on the angle between two vectors – it is the cosine of the angle
  - considers only the orientation of the vectors, not their magnitudes
  - 1 for angle of 0°, less than 1 for other angles
  - typically used only in positive space, so it is bounded in [0, 1]

$$L_C(A,B) = \frac{A \cdot B}{\|A\| * \|B\|} = \frac{\sum_{i=1}^{m} A_i * B_i}{\sqrt{\sum_{i=1}^{m}(A_i)^2} * \sqrt{\sum_{i=1}^{m}(B_i)^2}}$$

Similarity measures

- Euclidean distance
  - distance in the vector space

$$L_E(A,B) = \sqrt{\sum_{i=1}^{m} (A_i - B_i)^2}$$

- Jaccard coefficient (index)
  - the size of the intersection of two sets divided by the size of their union

$$L_J(A,B) = \frac{A \cdot B}{\|A\| + \|B\| - A \cdot B} =$$

$$= \frac{\sum_{i=1}^{m} A_i * B_i}{\sqrt{\sum_{i=1}^{m} (A_i)^2} + \sqrt{\sum_{i=1}^{m} (B_i)^2} - \sum_{i=1}^{m} A_i * B_i}$$

Evaluating the clustering process

- a correct result (labels) is not known unlike in a supervised learning problem
- the perfectness of the output is usually expected to be much lower that desired because only a human has a clear objective and can use some additional, external information
- a user typically wants a reasonable (low) number of clusters with sufficient quality
- approaches use
  - internal evaluation measures
  - external evaluation measures
  - expert opinion

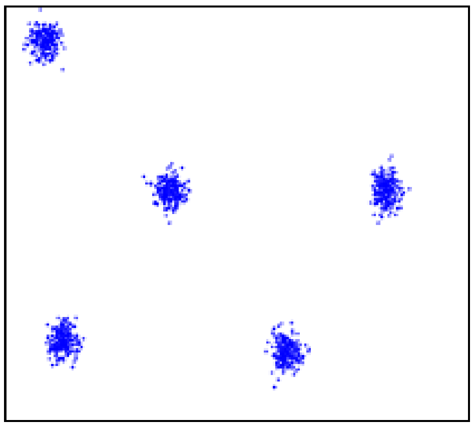
Internal evaluation measures

- derived from the data itself
- usually based on the criteria of
  - compactness
    - how much the objects are in a cluster related to each other (low variance = better compactness)
    - measures by maximum or average pairwise distance or maximum or average center-based distances
  - separation
    - how a cluster is separated from other clusters
    - measured by distances between cluster centers, minimum pairwise distances between objects from different clusters, or measures based on density

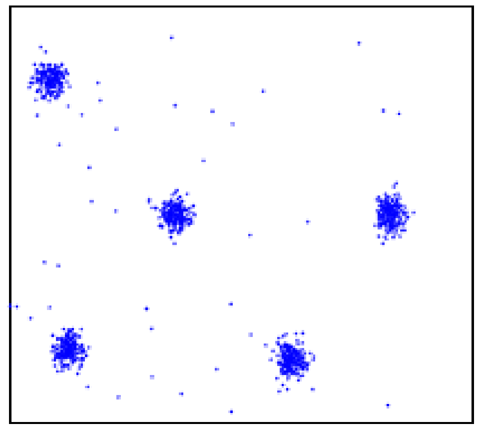
Internal evaluation measures optimal values

- depend on how their value changes with an increasing number of clusters
- optimal number of clusters at the function
  - elbow (knee) – for monotonic measures (Root-mean-square std. dev., R-squared, Modified Hubert Γ statistic)
  - maximum or minimum for the others (Dunn index, Silhouette index, …)

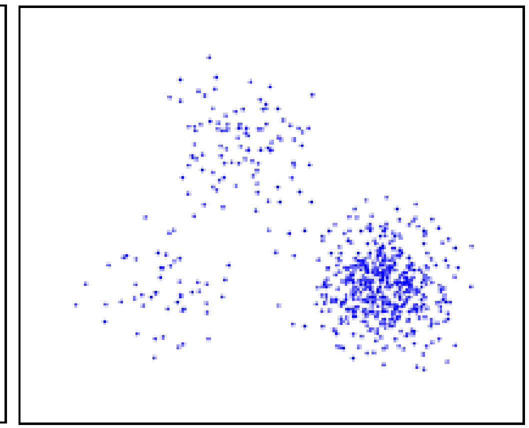
Internal evaluation measures – problems



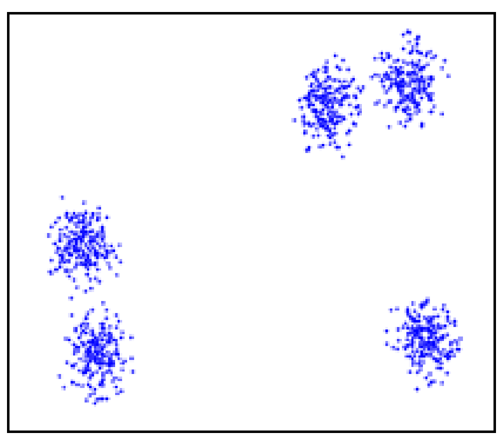
well separated



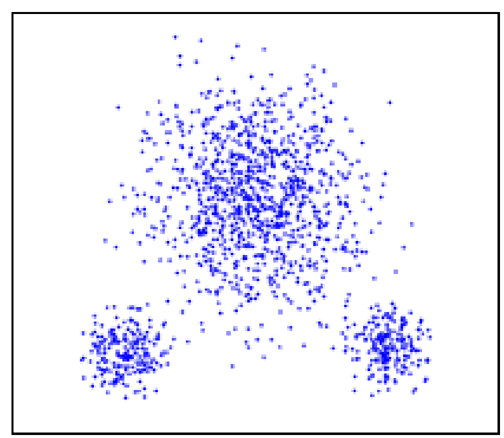
well separated with noise



different density



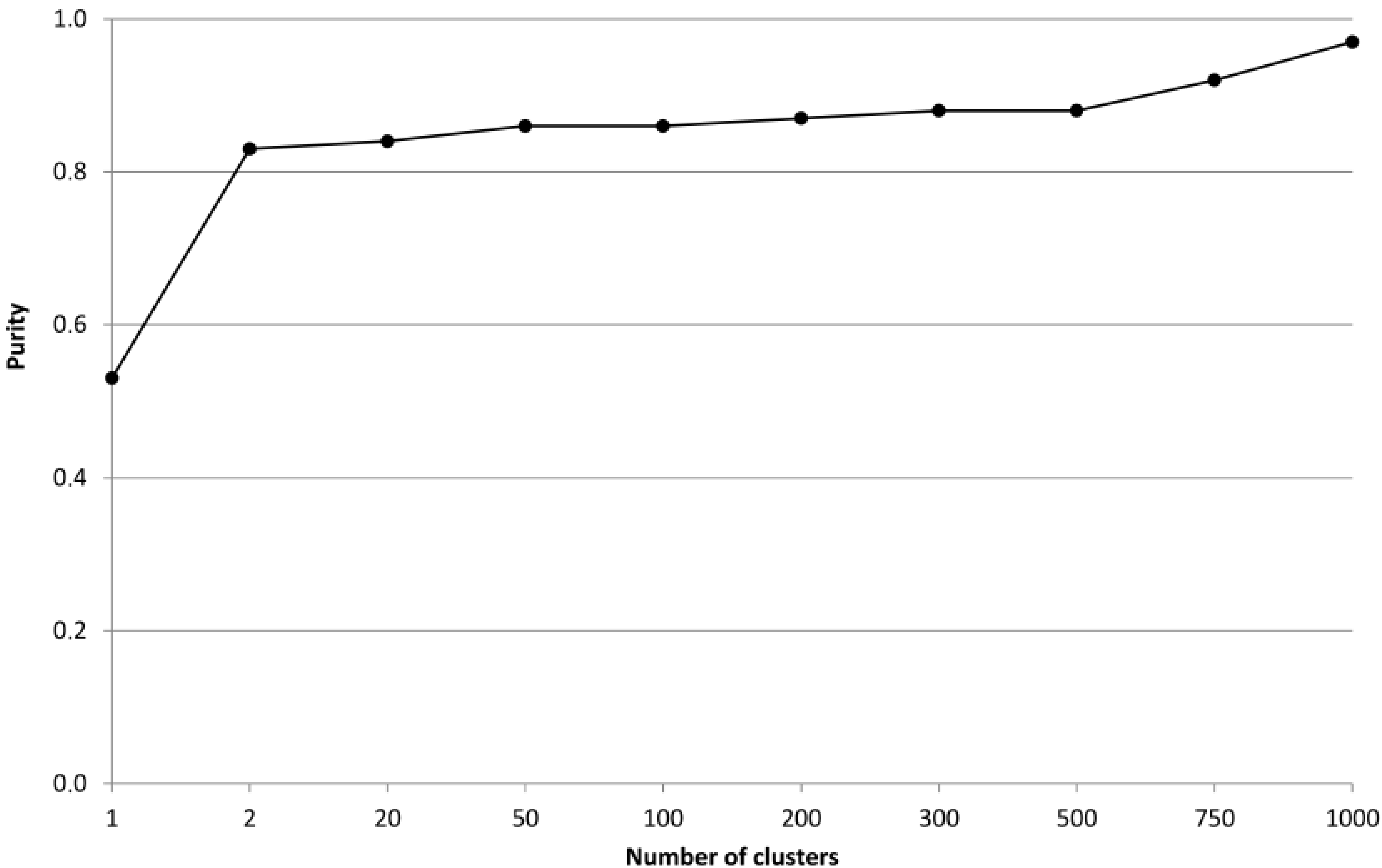
subclusters



skewed distribution

External evaluation measures

- class labels (ground truth) for the data to be clustered are available a priori
- entropy: $E_j = -\sum_i p_{ij} \log p_{ij}$ (p_{ij} – probability that a member of the cluster *j* belongs to the class *i*)
- purity: $P_j = \frac{1}{n_j} * \max_{i=1..l} n_{ij}$ (n_j is the number of instances in the cluster *j*, n_{ij} is the number of instances of the class *i* in the cluster *j*, and *l* is the number of classes)



Quality of clustering solutions with different numbers of clusters measured by the Purity criterion

Evaluation based on expert opinion

- suitable under many circumstances
- may reveal new insight into the data
- generally very expensive and demanding
- the results that are subjectively influenced are also not very well comparable

Evaluation based on expert opinion

- visual examination
  - not possible when the number of dimensions > 3
  - objects might be projected from a multidimensional space to 2D or 3D (groups of objects that are separated in 2D or 3D space are also separated in a space with many dimensions
  - difficult to use for data with a very high number of dimensions

Evaluation based on expert opinion

- examining the objects in individual clusters
  - usually not possible to consider not all of the documents but examine only some of them:
    - an average document – lies most closely to the remaining documents in the cluster (located near the centroid)
    - the most typical – located near the border of the cluster, far from the other clusters (the most different from all of the documents in the collection = most specific)
    - the least typical element – close the border of the cluster, closely to the remaining clusters; it is most similar to the documents in the other clusters

Evaluation based on expert opinion with machine learning support

- an advanced automated method that would help in evaluating individual clusters is useful
- relevant attributes that sufficiently well characterize the documents in the clusters might be filtered
- a feature selection method using clusters as classes is an option